

a post-rolling filter is applied to reduce visual noise and to make the comparison between measured and estimated SOH easier to interpret.

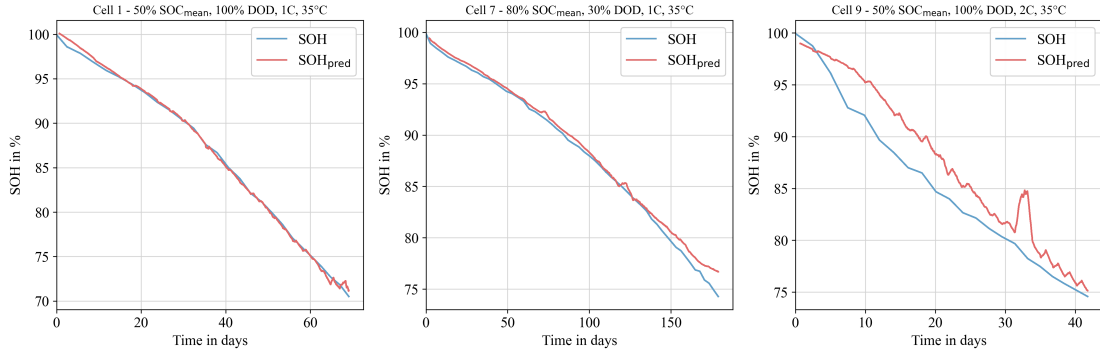


Figure 5.6: Comparison of predicted and measured SOH trajectories over time for three representative test cells. Cells 7 and 1 show the typical ageing progression most frequently observed in the dataset and are estimated accurately by the network, whereas Cell 9 was intentionally chosen as an atypical example with a pronounced U-shaped SOH curve. A post-rolling filter is applied to reduce visual noise and to make the agreement and deviation between measured and predicted trajectories easier to interpret.

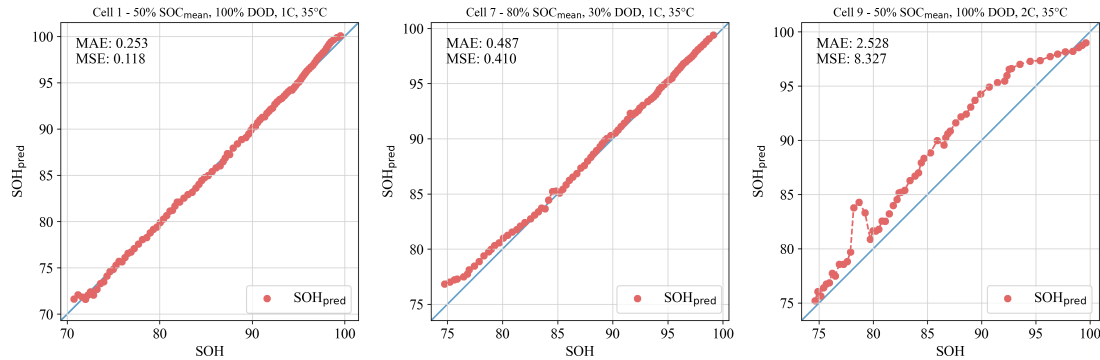


Figure 5.7: Scatter comparison between predicted and measured SOH values for the three representative test cells. Cells 7 and 1 remain close to the identity line and therefore exhibit low prediction errors. For Cell 9, the model follows parts of the qualitative SOH development, but the larger distances from the identity line show that the absolute SOH values are reproduced less accurately. The atypical trajectory alone does not cause the error. MAE and MSE quantify the residual deviations between prediction and reference. Error values are shown on the 0 to 100% SOH scale, with MAE in percentage points and MSE in squared percentage points.